

WITHANAGE DON CHATHUSHKA NAVOD

Software Engineer | Final-Year IT Undergraduate, SLIIT

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SUMMARY

Final-year Software Engineering undergraduate with hands-on production experience building full-stack, desktop, and AI-agent systems end to end — from a self-hosted multi-agent platform (LangGraph, MCP, n8n), to a 44-table Electron/React POS running live in retail tech shops and its cloud-native multi-tenant SaaS rebuild, to contributing on an LLM-powered Sinhala writing assistant. Comfortable across the stack: React/Next.js, Node.js, Python, PostgreSQL, and shipping to real infrastructure (Docker, CI/CD, VPS).

WORK EXPERIENCE

Full Stack Developer Intern

Dec 2024 – Jun 2025

Webminds, Malabe

- Built and maintained web applications using the MERN stack, Next.js, and TypeScript.
- Wrote and maintained Jest test suites to keep code quality high across feature branches.
- Handled deployments, server configuration, and continuous integration.

Sail Craft Technician

May 2021 – Oct 2022

North Sails, Biyagama

- Performed equipment inspection, maintenance, and inventory management in a technical manufacturing environment before transitioning into software.

FEATURED PROJECTS

Second Brain — Personal Multi-Agent AI Platform

Jun 2026 – Present

Python, LangGraph, FastAPI, Next.js, MongoDB, PostgreSQL, MCP, n8n

- Designing and building a self-hosted multi-agent AI system: a LangGraph orchestrator coordinating per-module FastMCP servers, a Postgres (Neon) "trust layer," MongoDB Atlas for module data, and a Next.js dashboard — deployed end-to-end on a self-managed VPS.
- Built a human-in-the-loop approval queue (Postgres-backed executor polling and running approved actions) with a code-enforced confidence hard-cap on destructive actions, decoupled from LLM judgment for safety.
- Shipped the first production module — automated job search and tracking — end to end, plus n8n automation workflows (job-match pipeline, YouTube content pipeline) wired into the agent infrastructure via a FastAPI webhook front door.
- Built the Next.js dashboard (approvals, activity, jobs, modules, settings) and hardened the VPS deployment with systemd, Nginx, and memory-capped services on a constrained 1GB production box.

Kodernet POS — Desktop (TechShop Edition)

May 2026 – Jul 2026

Electron, React 19, SQLite, Supabase

- Built and maintain a production Electron/React POS & ERP system for tech-repair retailers — 44 database tables, 25+ toggleable modules (billing, inventory, repairs, warranties, PC-build configurator, multi-branch transfers, HR/payroll), and 298 IPC channels between renderer and main process.
- Designed an offline-first data layer on SQLite (better-sqlite3, WAL mode) with a bidirectional outbox-pattern Supabase/Postgres sync engine for real-time multi-device replication.
- Implemented specialised retail workflows: serial/IMEI-tracked inventory with full lifecycle history, Kanban-based repair job sheets, warranty claim processing, and a PC-builder with a live compatibility-checking engine.
- Automated silent thermal receipt and Code128 barcode label printing via native OS print spoolers, built a sync bridge to a PHP/MySQL e-commerce storefront, and packaged the app as a signed Windows installer with license activation.

Kodernet Cloud POS — SaaS Rebuild

Aug 2026 – Present

Next.js, Node.js, PostgreSQL, Turborepo, Docker

- Leading the build of a multi-tenant cloud version of the desktop POS, architected as a modular monolith (schema-per-tenant Postgres) designed to run on a sub-\$30/month infrastructure budget with a defined extraction path to microservices at scale.
- Structured the codebase as a Turborepo monorepo (web, admin, and API apps plus shared contracts/UI packages) with GitHub Actions CI/CD and Terraform/Ansible-provisioned infrastructure, and implemented tenant provisioning, subscription/billing lifecycle, and an offline-first PWA billing flow.
- Authored the full architecture and requirements specification — business analysis, competitor analysis, infrastructure cost modelling, and security/ADR documentation — driving every build decision.

SinhalaJournalLLM — University Research Project (R26-SE-037)

Jul 2026 – Present

FastAPI, React, Supabase, LLM fine-tuning

- Contributing engineer on a 4-person team building a FastAPI + React + Supabase platform delivering four LLM-powered Sinhala writing tools (grammar correction, headline generation, style rewriting, summarisation) on fine-tuned SinLlama (Llama-3-8B) LoRA adapters, shipped as a web app, Chrome extension, and Google Docs add-on.
- Built a Model Comparison / Summariser Lab dashboard for benchmarking adapters side by side, and diagnosed and fixed inference-decoding bugs across the summarisation pipeline (prompt double-wrapping, empty-separator crashes, Sinhala Unicode text-healing regressions).
- Implemented Google Sign-In on the auth backend and built the Chrome extension and Google Docs add-on integrations; co-authoring the accompanying research paper on fine-tuning approaches for low-resource-language LLMs.

EDUCATION

BSc (Hons) Information Technology, Specialized in Software Engineering

2022 – Present (Final Year)

SLIIT, Malabe

Diploma in English Language

2021

Esoft Metro Campus

G.C.E. Advanced Level — Engineering Technology stream

2017 – 2020

Henegama Central College

TECHNICAL SKILLS

Languages: JavaScript/TypeScript, Python, Java, C/C++, PHP

Frontend: React, Next.js, Flutter, Tailwind CSS, HTML/CSS

Backend & Platforms: Node.js, Express, Electron, FastAPI, Laravel

Data: PostgreSQL, MySQL, MongoDB, SQLite, Supabase

Tools & Infra: Git/GitHub, Docker, GitHub Actions CI/CD, Nginx, Linux VPS, Jest, Figma

AI & Agents: LangChain/LangGraph, MCP (Model Context Protocol), LLM Integration, Prompt Engineering, n8n

SOFT SKILLS

Problem solving • Team collaboration • Time management • Adaptability • Communication

REFERENCES

Mr. M C Sampath — Director, Hayles Fiber | 0765391746 | chaminda@haylesfiber.com

Mr. Anuja Vithanage — Manager, Printcare Secure Ltd | 0719524132 | anujahv@gmail.com